

Fares Ben Slimane

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SUMMARY

Senior AI Engineer specializing in end-to-end AI product delivery, from problem definition to production-grade deployment. Proven experience applying AI across diverse use cases (Agriculture, Finance, Healthcare, Industry..etc). Client-facing and value-driven, with a strong ability to translate business and operational pain points into actionable AI solutions. Comfortable working in Agile environments, collaborating closely with stakeholders to deliver measurable, production-ready impact.

TECHNICAL SKILLS

Programming Languages: Python, C++, SQL, Javascript

Deep Learning Frameworks: TensorFlow, Keras, PyTorch (preferred)

Machine learning: Data analysis & visualization (Matplotlib, Plotly..etc), Supervised classification and regression ML algorithms, unsupervised ML algorithms, Statistics, Probability, Calculus and Linear algebra

MLOps & Cloud: Azure Cloud, Infrastructure as Code (Terraform, Duplo), Git, CI/CD (GitHub Actions), Containers & Orchestration (Docker, Kubernetes), Data Orchestration (Prefect), MLflow, Model Serving & APIs (FastAPI, Dash)

Libraries & Tools: NumPy, Pandas, Scikit-learn, OpenCV, TensorRT, ONNX

Deep learning (Theoretical & Practical): Neural Networks & Convolutional neural networks (CNNs), Recurrent Networks (RNN, GRU and LSTM), Transformers (attention-based models), Generative Models (GAN, VAE), Reinforcement Learning (Q-learning, Monte Carlo, Sarsa, PPO..etc)

NLP: OpenAI, LLM, RAG, Text classification, Language modelling, Machine translation

Computer Vision: Image classification, segmentation, object detection, anomaly detection, Video and Image Analysis

Speech Recognition: Lightweight Wakeword Detection, Command detection, ASR, Speech-to-Text and Text-to-Speech

Soft Skills: High-paced startup environments with lean teams working on R&D ML projects. Team player with efficient communication skills. Positive force and a good motivator. Effective in both spoken and written English and French

EXPERIENCE

[Airudi] Senior AI Engineer

January 2025 – Present

Healthcare, MLOps

Montreal

- Delivered production AI solutions in the healthcare domain, driving measurable operational impact for large-scale public services in Quebec.
- Client-facing role with ownership of the full MLOps lifecycle, from problem definition and solution scoping to deployment, monitoring, and continuous improvement.

[OpenClassrooms] AI Mentor

October 2023 – Present

ML Development, AI Project Management

Remote

- Mentored professional learners on core AI topics including data analysis, visualization, regression, classification, clustering, and AI project management.

[Fluent.ai] Machine Learning Developer

June 2022 – Feb 2024

Speech Recognition, Wake Word Detection, MLOps

Montreal, Canada

- Improved end-to-end data pipelines, including data collection, cleaning, and preparation, in collaboration with data scientists and engineers.
- Maintained and optimized the full MLOps lifecycle for wake-word models, covering training, evaluation, logging, deployment, monitoring, and iteration in production.
- Ported and integrated ML models across embedded DSP platforms, including HiFi 4/5 and Syntiant.
- Designed a wake-word architecture reducing model size by 55% and FLOPs by 16%, while achieving 7% FRR and 2–3 false alarms per wake word in noisy, multi-accent environments.

[Hummingbirds AI] Machine Learning Developer*Computer Vision, Biometrics*

December 2020 – May 2022

Remote

- Implemented state-of-the-art computer vision algorithms for object detection, segmentation, and multi-camera tracking.
- Engineered a robust person-tracking system handling occlusions and cross-camera identity association.
- Deployed a low-latency biometric system for face identification with strong anti-spoofing protection against 2D and 3D attacks.
- Collaborated directly with clients to translate business needs into actionable AI solutions.

[Ciena] R&D Computer Vision Developer*Visual Inspection, Anomaly Detection*

September 2019 – September 2020

Ottawa, Canada

- Built an automated visual inspection pipeline for PCB fault detection to support manufacturing quality control.
- Developed component detection algorithms for accurate identification of PCB elements.
- Applied unsupervised anomaly detection techniques to identify defects in PCB cards.

PROJECTS & PUBLICATIONS

Conference Paper: ICPR*Context Matters: Self-Attention for Sign Language Recognition*

Accepted

2020

- Arxiv Link: arxiv.org/abs/2101.04632

Mila Course (Prof. Aaron Courville)*Representation Learning (IFT 6135)*

Grade: A-

Winter-2019

- Linear algebra, Calculus and optimization in Deep Learning
- Neural Networks, Recurrent Networks, CNN, Generative Models theory
- Developing DL algorithms and applying them in real use cases

Sign Language Recognition & Translation

2019

- Build a system that interprets a sequence of images, representing sign language, and generates a coherent textual translation in spoken language. Implemented advanced capabilities to effectively learn and extract essential spatio-temporal information from sign gestures, ensuring accurate and meaningful translations.

[Orange] R&D Machine Learning Developer

February 2017 – June 2017

Tunis, Tunisia

- Designed and constructed a prototype for an intelligent hydroponic farm for plants.
- Implemented an AI rule-based system to manage indoor settings autonomously.
- Employed Deep learning Computer vision models to identify plant anomalies and diseases based on leaf appearance.
- Established real-time control and monitoring of internal farm parameters through an intuitive web dashboard.

EDUCATION

M.S. in Computer Science (Artificial Intelligence)

GPA: 3.94

University of Quebec at Montreal

Jan 2018 - Sep 2020

Higher institute of information and communication technologies (ISTIC)*'Licence' in Computer science*

Tunis, Tunisia

Sep 2014 - Jun 2017

AWARDS

Scholarship of Excellence (UQAM)*Faculty of Science - MSc Computer Science*

Montreal, Canada

2018 - 2019

Scholarship Mitacs Accelerate*Mitacs Accelerate Program*

Montreal, Canada

2019 - 2020